

GRACE M. INMAN, EIT

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M.S. student in Civil & Environmental Engineering with three years of professional water resources experience. Focus on the software side of water projects: geospatial Python, data pipelines, and dashboards. Experience with managing field crews, scheduling, and project logistics. Graduating May 2027.

EDUCATION

Duke University — M.S., Civil & Environmental Engineering

Expected May 2027

Concentration in Systems, Risk, and Decision. Graduate research: estimating rainfall-derived inflow and infiltration (RDII) in sewer systems without relying on rainfall records.

Relevant coursework: High-Performance Scientific Computing; Water Law and Policy; Multi-objective Optimization; Spatial Data Analysis; Computer Science Fundamentals.

Tufts University — B.S., Environmental Engineering

2023

Relevant coursework: Urban Hydrology; Data Science and Machine Learning; Statistics and Numerical Methods; Engineering Management.

EXPERIENCE

River Ranger (Intern) — Student Conservation Association / U.S. Army Corps of Engineers

Summer 2026

- Support district operations and deliver public water safety outreach at locks and dams on the Allegheny, Monongahela, and Ohio Rivers.

GIS Developer and Field Team Lead — Edible Atlas (formerly David Lab), Duke University

2025 – Present

- Build and maintain the lab's community wastewater eDNA dashboard: a Python/Dash application deployed on Render, serving map, site, diet, richness, and rainfall views over a DuckDB and Parquet data store.
- Own the staged R-to-Python ETL pipeline behind it, converting raw sequencing output, Excel, and shapefile inputs into the dashboard's processed datasets.
- Maintain the lab's GIS layers and service-area geospatial data; currently scoping new sampling sites.
- Manage training, scheduling, and logistics for a monthly sewer sampling program across 25 census tracts in Durham, NC, for a field crew that grew from seven to ten technicians.

Water Resource Engineer I/II — CDM Smith

2022 – 2025

- Served as task manager for a regulator structure manual documenting weirs and other sewer regulator structures for the City of New Bedford, MA, coordinating interns and GIS specialists; integrated CAD drawings, GIS data, and 3D scans into the final manual.
- Supported development of the Old Colony Regional Water Plan, researching regulatory frameworks and evaluating supply-demand scenarios for 17 municipalities in southern Massachusetts.
- Performed infiltration and inflow (I/I) analyses for a Sanitary Sewer Evaluation Study (SSES) for the Boston Water and Sewer Commission, using EPA SSOAP to identify sources of extraneous flow.
- Developed hydrologic and hydraulic models for FEMA riverine flood insurance studies, including 2D HEC-RAS floodplain mapping and geospatial processing in ArcGIS and HEC-HMS.
- Supported water quality and hydraulic modeling for the Allegheny County Sanitary Authority (ALCOSAN), automating preprocessing and post-processing of EPA SWMM and EFDC model results.
- Conducted wet- and dry-weather field sampling for a water quality improvement study at Sassaquin Pond (New Bedford, MA) and established both an automated sampling program and a volunteer monitoring program to characterize nutrient and bacteria impairments.

Music Event Coordinator — Tufts University

2019 – 2023

- Coordinated performers, audiovisual staff, and venue operations for live music events; promoted from Music Event Staffer based on performance and leadership.

Data Laboratory Assistant — Tufts University

2021 – 2022

- Provided GIS, data visualization, and coding support to students and faculty, and maintained the Data Lab space.

Science Undergraduate Laboratory Intern (SULI) — Ames Laboratory, U.S. Dept. of Energy

Fall 2020

- Researched chemical recycling methods for end-of-life magnets to recover the rare-earth elements neodymium, dysprosium, and praseodymium used in electric vehicle production.
- First author on two peer-reviewed publications; presented findings at the Critical Materials Institute 2021 Winter Meeting and the 2021 American Chemical Society annual conference.

Research Assistant — Environmental and Health Nanoscience Lab, Brown University

Summer 2019

- Characterized and functionalized graphene oxide hydrogels for use as aquatic passive samplers for PFAS, under the NIEHS Superfund Research Program.

SELECTED PROJECTS

Sewershed delineation tool — Streamlit application that traces the sewer network upstream of a given manhole and returns the contributing service-area polygon plus census-based demographic estimates of the population served. Built in Python using graph traversal over the network topology and spatial joins against census geography; used by lab members to scope and interpret sampling sites.

RDII analysis pipeline (graduate research) — Python pipeline processing sewer flow meters into basin-level rainfall-derived inflow and infiltration indices that inform capital improvement planning for sewer rehabilitation.

SKILLS

Programming: Python (primary), R, MATLAB, SQL

Computing: software engineering practice (refactoring, unit testing, profiling, packaging), databases, command line and bash

Applied areas: geospatial analysis, data pipelines and ETL, interactive dashboards, statistical modeling and machine learning

Water modeling: EPA SWMM, EPA SSOAP, HEC-RAS, HEC-HMS, EFDC

GIS and spatial: ArcGIS, ArcGIS Online, geospatial Python, GIS field tools

Workflow and deployment: Git and GitHub, conda, Render, Linear, Bluebeam Revu

PUBLICATIONS

Inman, G., Prodius, D., and Nlebedim, I.C. "Recent Advances in Acid-Free Dissolution and Separation of Rare-Earth Elements from Magnet Waste." *Clean Technologies and Recycling*, November 2021.

Inman, G., Nlebedim, I.C., and Prodius, D. "Application of Ionic Liquids for the Recycling and Recovery of Technologically Critical and Valuable Metals." *Energies*, January 2022.

CERTIFICATIONS, AWARDS, AND AFFILIATIONS

WEF InFLOW STEMpath Scholar, 2026 — selected for the Water Environment Federation's national program introducing students to the water sector, including sponsored attendance at WEFTEC.

Engineer-in-Training (EIT), Massachusetts

Water Environment Federation (WEF); New England Water Environment Association (NEWEA)